Assignment (02/03/2026)

Assignment Name : ML Idea Generator

Description

Suggest machine learning problems in college, healthcare, and shopping and explain the **input → output** of each system.

Introduction

Machine Learning is widely used today to solve real-life problems by analyzing data and identifying patterns. It helps in making predictions, recommendations, and decisions automatically. Many fields such as education, healthcare, and online shopping can benefit from machine learning systems that make processes faster and smarter.

1. College Domain

Problem: Student Performance Prediction

Input

- Study hours
- Attendance percentage
- Previous semester marks
- Assignment scores

Output

- Predicted final exam marks or grade

Explanation

In colleges, teachers often want to know which students may need extra help. A machine learning model can analyze factors such as study hours, attendance, and previous marks to predict a student's final performance. This can help teachers support students who might be struggling.

2. Healthcare Domain

Problem : Disease Prediction System

Input

- Patient age
- Symptoms
- Blood pressure
- Sugar level
- Medical history

Output

- Predicted disease or possible health risk

Explanation

In healthcare, early detection of diseases is very important. A machine learning system can analyze patient data and symptoms to predict the possibility of certain diseases. This can help doctors make faster decisions and begin treatment earlier.

3. Shopping Domain

Problem : Product Recommendation System

Input

- User purchase history
- Product ratings
- Browsing history
- Customer preferences

Output

- Personalized product recommendations

Explanation

Online shopping platforms often suggest products to users based on their interests. A machine learning model analyzes what a customer has searched for, viewed, or purchased before and recommends similar or related products. This improves the shopping experience and helps users find items easily.

4. Smart Library System (College)

Problem : Book Recommendation System

Input

- Student reading history
- Preferred genres
- Book ratings

Output

- Recommended books for the student

Explanation

A smart library system can recommend books based on a student's reading interests and preferences. By analyzing previous books read and their genres, the system can suggest new books that the student might enjoy.

Conclusion

Machine Learning has the potential to improve many areas of daily life. In education, it can help predict student performance. In healthcare, it can assist in early disease detection. In shopping and libraries, it can provide personalized recommendations. These systems use data to generate useful insights and make processes more efficient.